SCIENCE DEPARTMENT • SCIENCE LITERACY & LESSON INTERNALIZATION

PD AGENDA

Strengthening Science Literacy: Calibrating Writing Benchmarks, Feedback, and Revision

Strategy Focus: Teaching Students to Analyze Science Sources With CUBS + CER — adapt a student-facing path from source analysis to an evidence-based explanation, feedback, and revision.

WEDNESDAY, OCTOBER 28, 2026 • 2:30–3:30 P.M. **60 MINUTES** **GRADES 9–12 SCIENCE • BRING FORMS & EVIDENCE**

TODAY'S OUTCOME Leave with a ready-to-teach CUBS/CER routine and a Lesson Planning & Reflection Sheet that identifies the student analysis strategy, the priority need shown by evidence, the feedback/revision plan, and the next evidence check.

WHY THIS PLC MATTERS Every science course asks students to interpret information and explain scientific thinking. CUBS and color-coded CER make source analysis, evidence use, and reasoning visible while students remain responsible for the thinking.

PROTOCOL CONNECTION
Every Day Questions 1, 3, and 4; Team Deep Dive Steps 2–4: clarify the learning goal and mastery bar, anticipate student responses and access needs, and plan the student experience, feedback, and revision moves.

WHAT TO BRING
One course-specific source or response task; 1–2 de-identified writing samples; current standard(s); student-facing materials; CER-color highlighters; Lesson Internalization Tool; and planning materials.

FOCUS TO PROTECT
The Biology benchmark is available only as an optional model. Every science teacher adapts CUBS and color-coded CER to an authentic course source and task. The routine supports students in doing the task; it is not an additional grading system.

TODAY'S PLC PLAN

TIME (MIN)	FOCUS	WHAT TO DO	PRODUCT / CHECK
2:30–2:41	**Frame the science literacy cycle**	Connect Monday's task experience and Tuesday's planning to the course-wide process: analyze a source, construct an evidence-based explanation, use feedback, and revise.	**Purpose + learning arc**
2:41–3:00	**Model CUBS + practice a course source**	Think aloud CUBS moves, then apply them to an authentic article, data display, diagram, model, investigation result, or task. The Biology benchmark is optional only.	**Source analysis**
3:00–3:15	**Color-code CER + plan feedback/revision**	Use a de-identified writing sample to make claim, evidence, reasoning, and linking language visible; plan a prompt and revision action tied to the mastery bar.	**Response + feedback**
3:15–3:22	**Teacher reading + writing strategy share**	Two or three teachers share another strategy students use to read/analyze a science article/source or write an explanation; discuss what works and any challenge. Teachers not using a strategy yet choose a CUBS/CER move from this PD to try next.	**Colleague learning**
3:22–3:28	**Course-alike lesson planning + reflection**	Use the Lesson Planning & Reflection Sheet to record the standard, source/task, student analysis strategy, data-based priority need, feedback/revision action, and next step.	**Ready-to-teach plan**
3:28–3:30	**Close + evidence plan**	Teach the routine, complete the after-lesson reflection section, record the evidence to collect, and bring a small de-identified set of completed writing and feedback/revision evidence Friday.	**Friday bridge**

PROTECTED INDIVIDUAL WORK WITH COLLEAGUE SUPPORT

WORK WITH COLLEAGUES

WHAT TO DO Use the Lesson Planning & Reflection Sheet to identify what students will analyze, the strategy they will use, the data-based need the lesson will address, the feedback/revision action, and the evidence check. Share a colleague idea or a strategy to try next.

1. **Select your source + task:** Use a text, graph, data display, diagram, model, investigation result, or scenario.
2. **Plan student analysis:** Choose the student-facing source-analysis strategy and exact directions.
3. **Name the priority need:** Use data or student work to identify the most important need to address.
4. **Plan feedback + revision:** Write ready-to-use directions, a feedback prompt, and a revision action.
5. **Reflect + adjust:** After the lesson, record what evidence shows, a colleague idea, and Friday evidence.

USE THIS REFERENCE DURING GROUP WORK

ADAPT THE ROUTINE TO YOUR SCIENCE COURSE
Begin with an authentic source and response task. Clarify what students must notice, explain, and make visible in a complete response.

STUDENT STRATEGIES MAKE THINKING VISIBLE
Students use a routine such as CUBS, diagram annotation, graph notice/wonder, or data-table analysis before writing an evidence-based explanation.

FEEDBACK MUST LEAD TO REVISION
Use a criterion-based feedback prompt that leads to a specific student revision. After the lesson, use evidence to decide what to keep, adjust, or try next.

COLLABORATION IN PRACTICE Protect time for thoughtful individual work; use respectful, student-centered feedback to make the next decision clearer.

TEACHER RUBRIC CONNECTION **Standard I: I-A-4 Well-Structured Lessons • Standard II: II-A-2 Student Engagement • Standard II: II-D-2 High Expectations • Standard IV: IV-C-1 Professional Collaboration**

